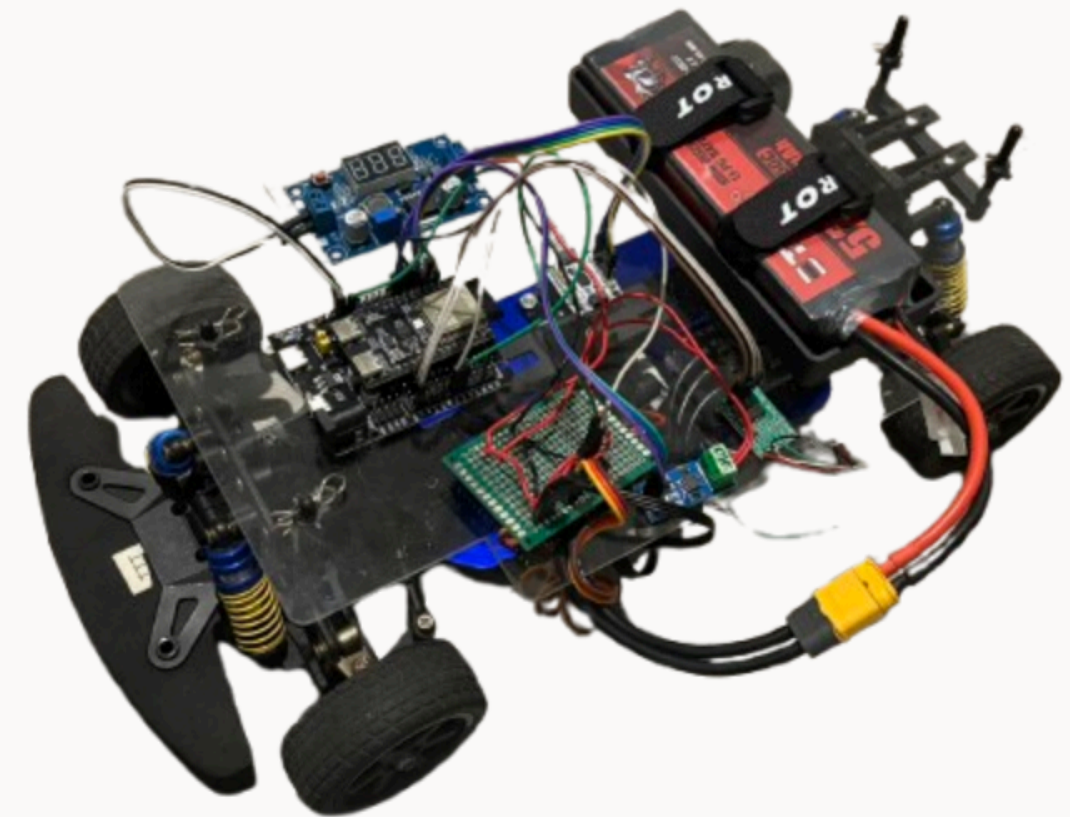




Force feedback in RC car scale



Presentation By: Bharuj Aursudkit

Problem Statement

What is steering wheel force feedback

Force Feedback (FFB) transmits reaction forces from the car to the driver through the steering wheel. Which gives the vehicle behavior to the driver

Problem

Most RC cars usually use Pistol-grip transmitters

- ✗ Lack physical feedback to the driver
- ✗ Driver receives only visual information
- ✗ Unrealistic driving experience
- ✗ Cannot feel vehicle behavior directly





Objective & Scope

Target Outcomes

- Latency < 70 ms
- Wireless Range $\geq$ 20 meters
- Force Response – Variable based on vehicle behavior

Scope

- Design FFB system on Sim Racing Wheel
- Use IMU (BNO086) and Load Cell to measure vehicle state
- Compute FFB on PC in Real-time



Force Feedback Formula

$$\tau_{FFB} = B \cdot \dot{\theta} + T_{fric} + K_{stiff} \cdot \theta + \tau_{SAT} + \tau_{Gravity} + \tau_{Surface_Jolt}$$

$B \cdot \dot{\theta}$	Damping – Resists fast steering motion
T_{fric}	Friction – Resistance force based on the vehicle speed
$K_{stiff} \cdot \theta$	Stiffness – Resistance from steering angle
τ_{SAT}	Self-Aligning Torque – Returns wheel to center
$\tau_{Gravity}$	Gravity – Compensates for vehicle tilt
$\tau_{Surface_Jolt}$	Surface Jolt – Road vibration feedback

Reference: Balachandran et al. (2014)



Load Cell

Why Add It?

- Reference: Karimi & Mann (2009) — "Torque feedback on the steering wheel of agricultural vehicles"
- Found that steering system resistance is part of "road feel"
- Helps driver perceive vehicle state better

How It Works

- Measures actual physical resistance in the steering linkage
- Not estimated — directly measured from the mechanism
- Added to passive force



Drift Detection

Why Add It?

- Reference: Wang et al. (2021) — "Remote driving testbed with force feedback based on slip angle estimation"
- Detects lateral slip to simulate loss of grip

How It Works

- If $|accY| > 3.0 \times (1 + |v|)$, reduce active FFB to 20%
- Simulates the "light" feeling when tires lose traction



Virtual Wheel Concept

Reaction forces
divided into 2 types:

1. Passive Resistive Forces

- Damping
- Friction
- Stiffness
- Load Cell

2. Active Restorative Forces

- Self-Aligning Torque (SAT)
- Gravity
- Surface Jolt
- Drift Detection

“Passive Force will only active when the wheel detect steering motion(steering velocity)”

Reference: Balachandran et al. (2014)

Hardware

Sensor

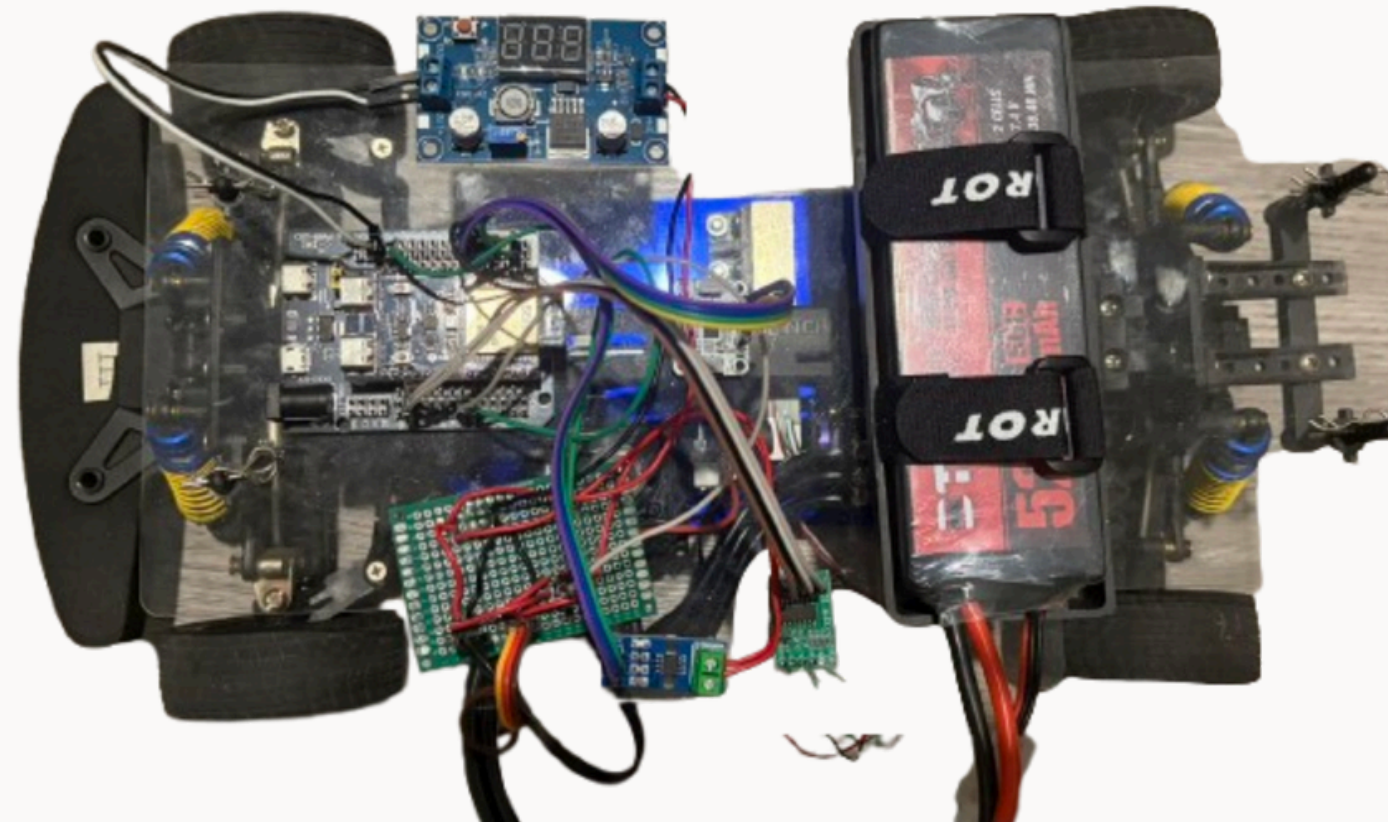
- Current Sensor (ACS712)
- IMU (BNO086 V2)
- Load Cell 1kg with HX711

RC Car

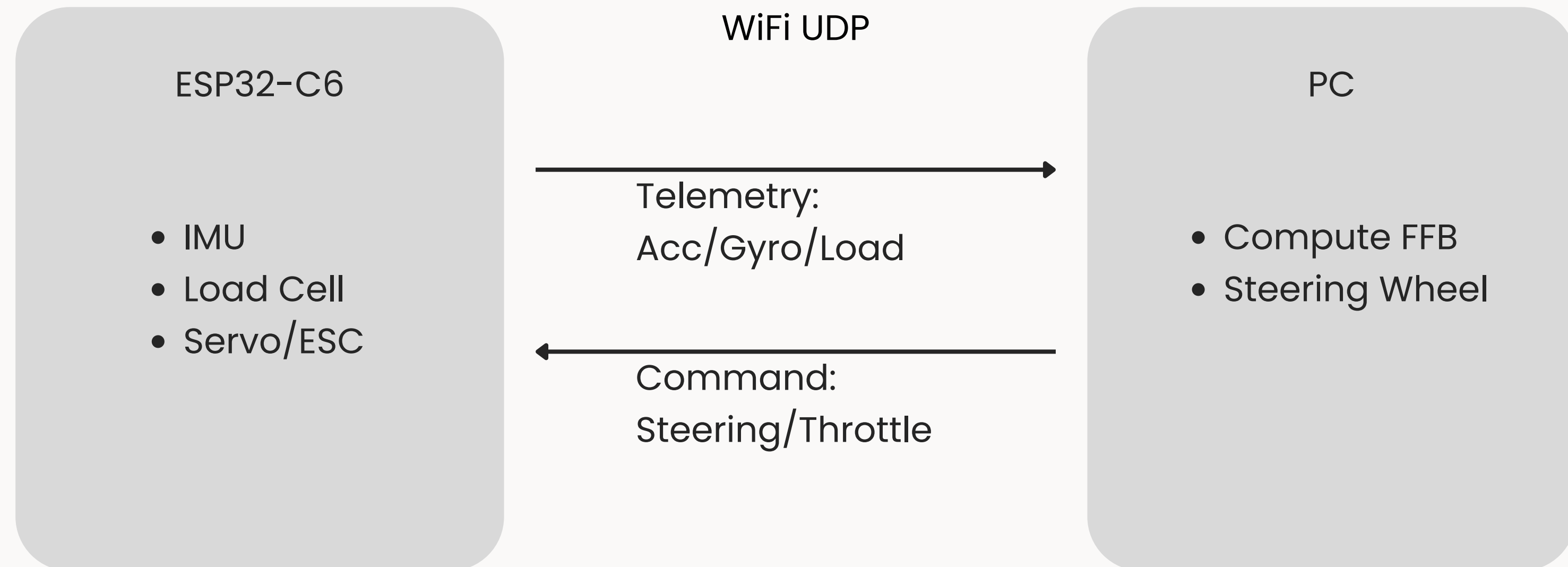
- ESP32-C6
- 1/10 Scale Chassis
- Rocket RC BLDC 10.5T
- Rocket RC 130A
- 9IMod 20 kg servo
- 2S LiPo 5200mAh
- LM2596

Steering Wheel

- Thrustmaster T248
- Thrustmaster T3PM



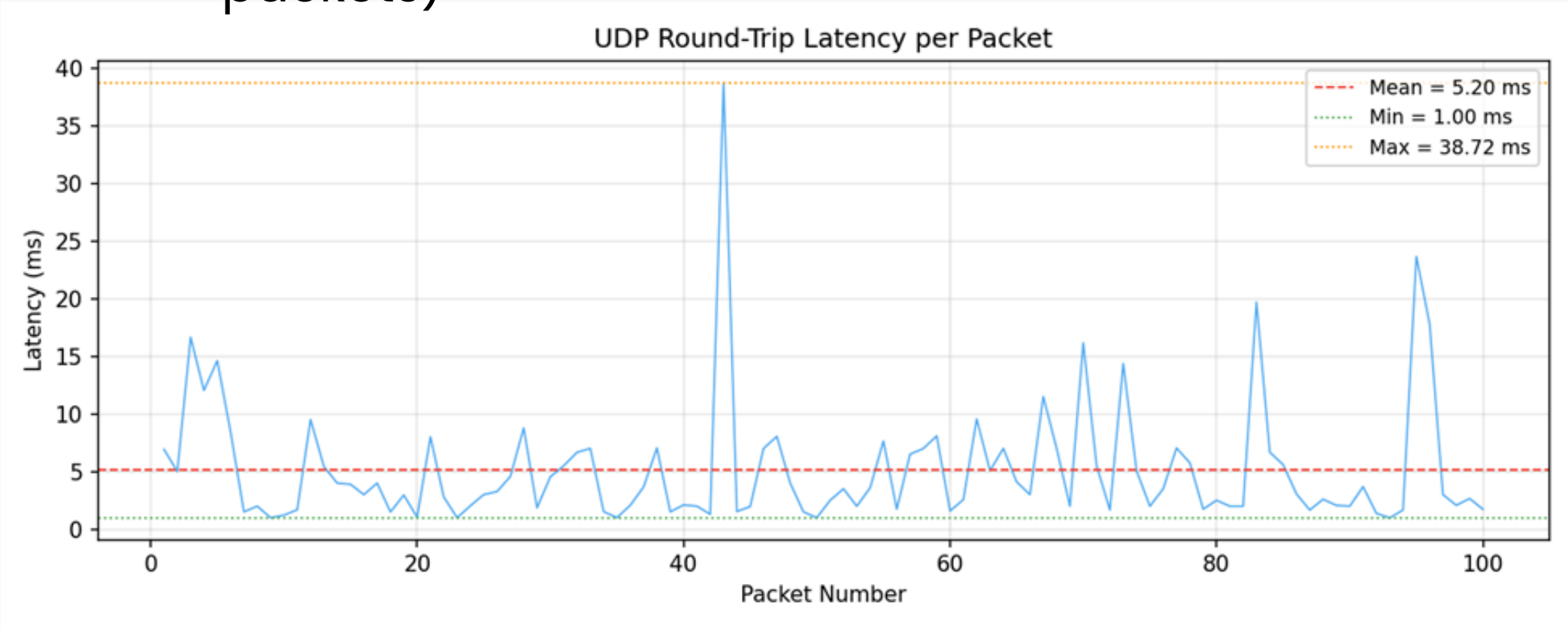
System Architecture





Communication Test Results

Latency Test (100 packets)

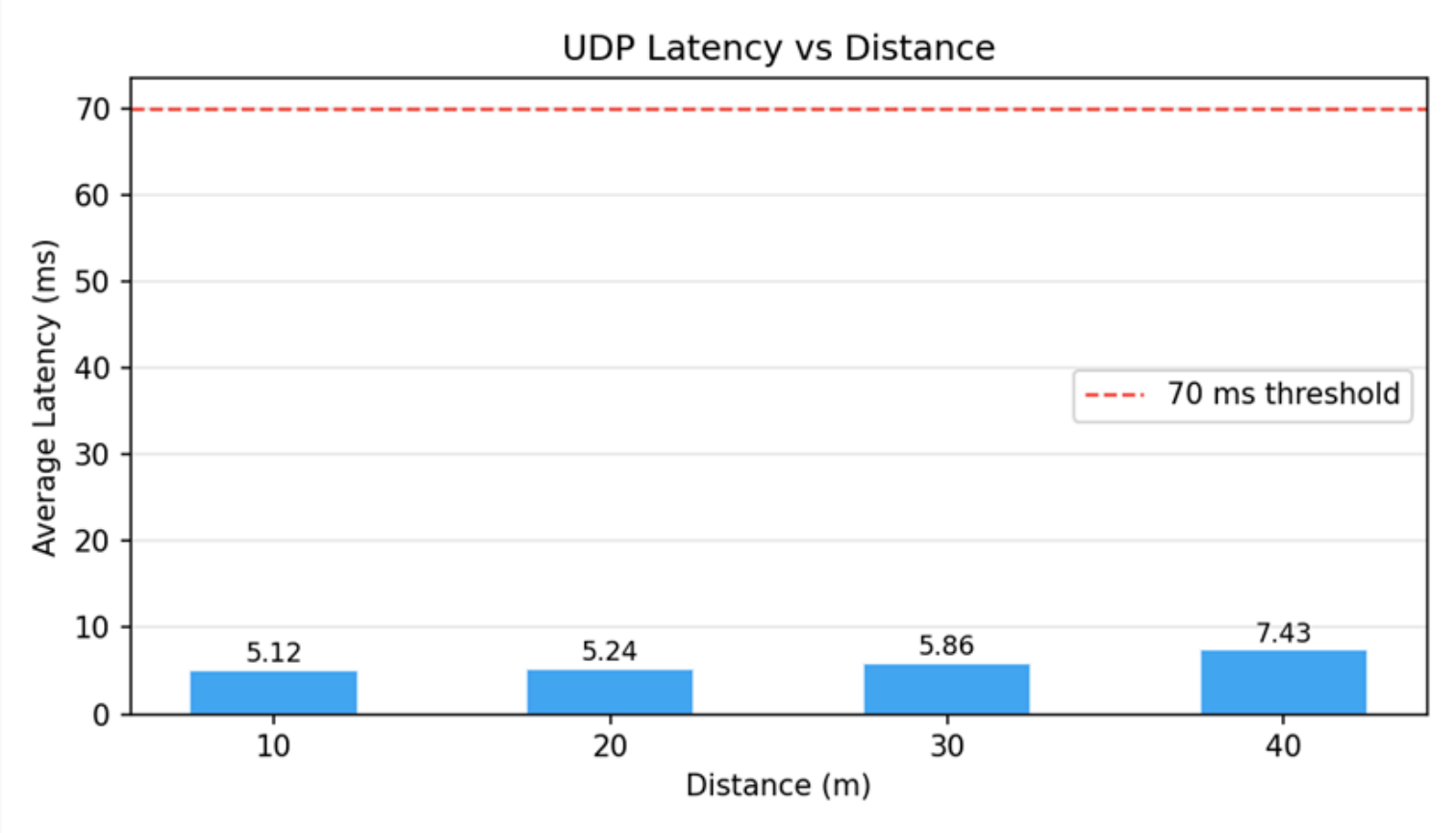


Parameter	Value
Mean	5.20 ms
Median	3.18 ms
Min	1 ms
Max	38.72 ms
Packet Loss	0.00%



Communication Test Results

Range Test



Distance	Latency	Packet Loss
10m	5.12 ms	0%
20m	5.24 ms	0%
30m	5.86 ms	0%
40m	7.43 ms	0%

Sensor Test Results

ACS712 Current Sensor

Status	Average	S.D.
Idle	2.011 A	0.120 A
Sweeping (moving)	2.140 A	0.255 A
Sweeping (Stationary)	2.032 A	0.143 A

- ✗ High noise (10% of Full Scale)
- ✗ Idle/Moving overlap makes separation hard





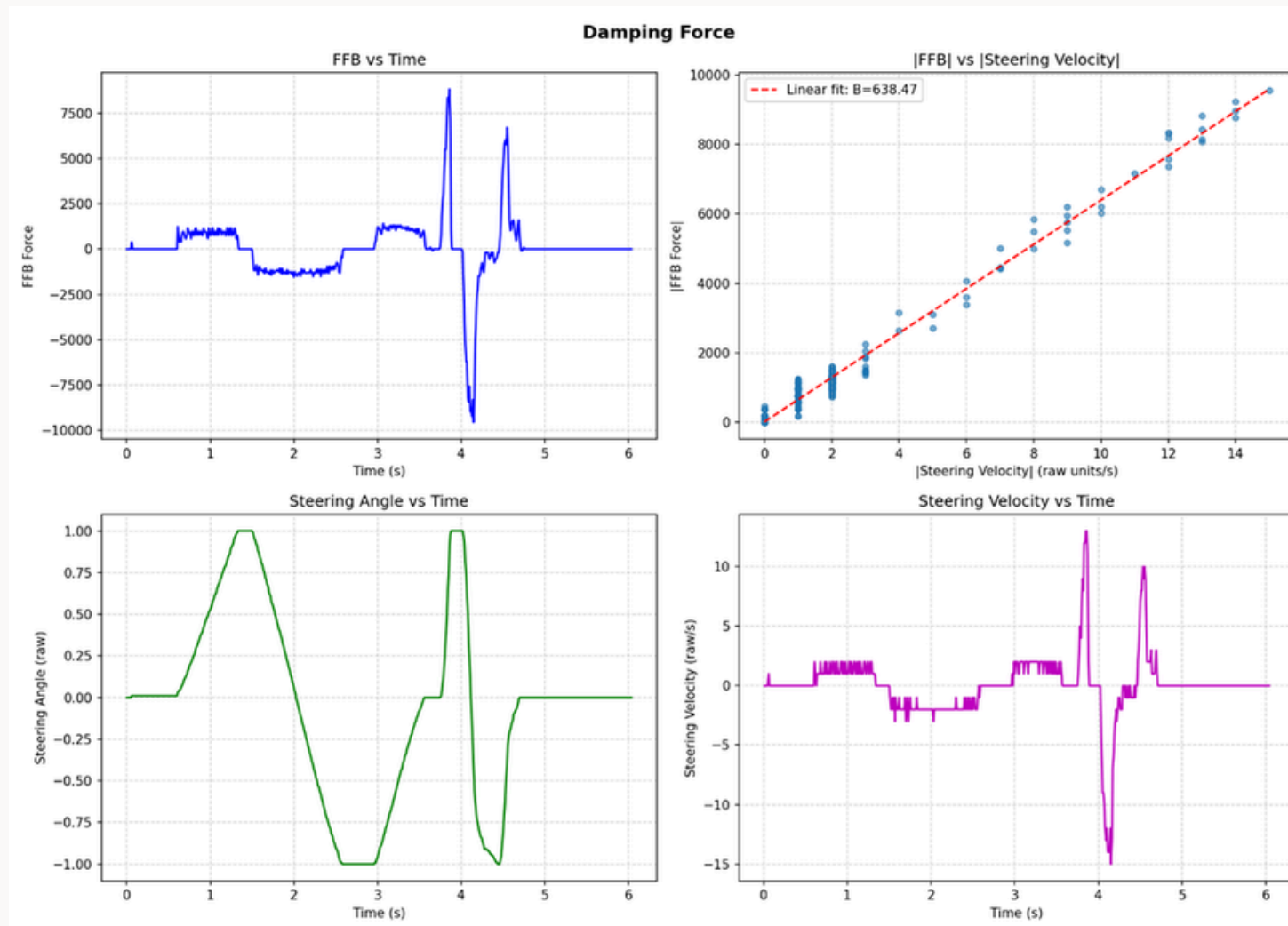
Force Response Test

FFB Result can be validated by individually testing each component of the FFB and evaluate the trend of input and FFB

Reference: Mandhata et al. (2012)

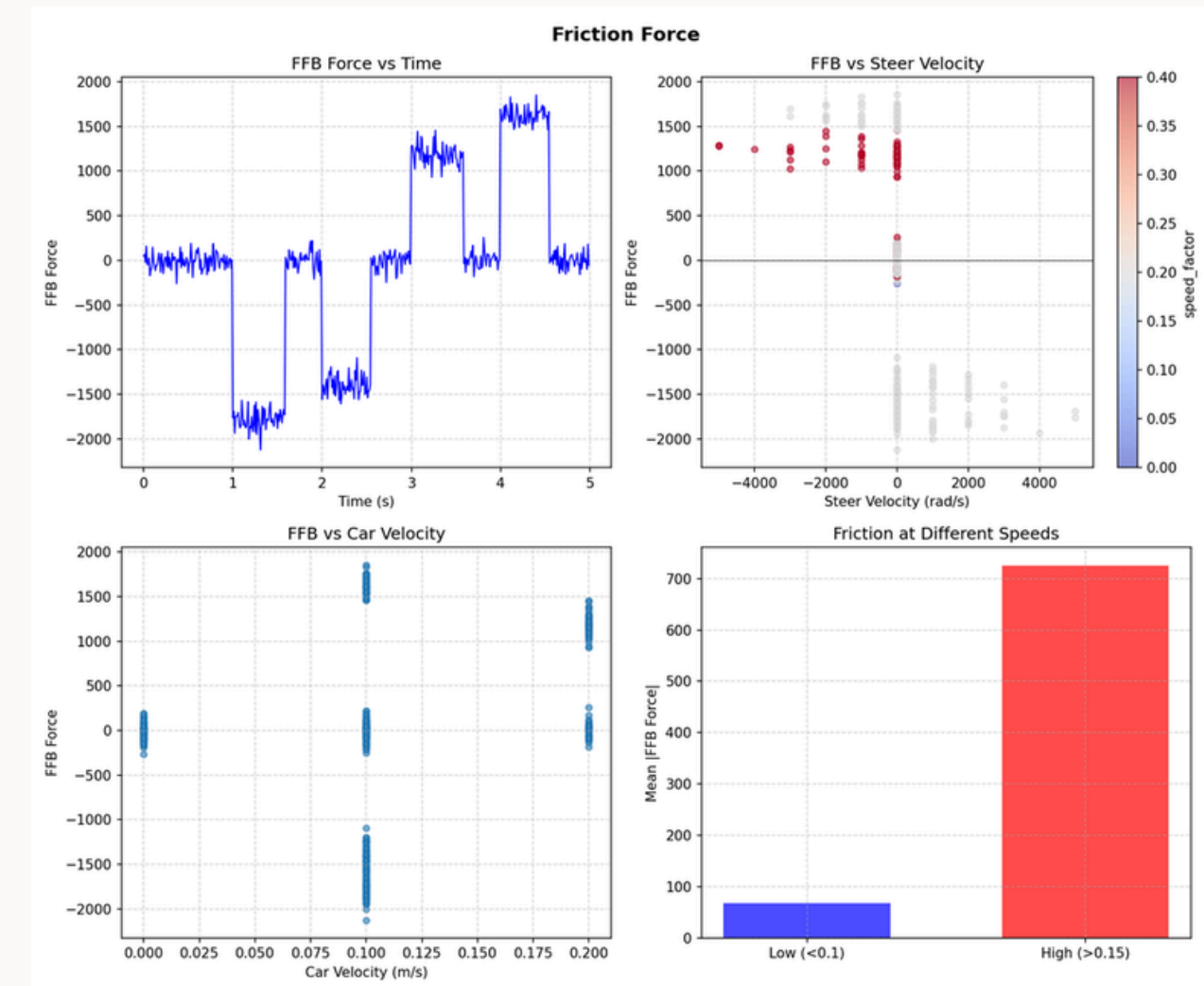
FFB Component Test Results

Damping Force



- Expected: $FFB \propto |\dot{\theta}|$ (linear with steering velocity)
- Result: $B \approx 638$, $R^2 \approx 0.95$ — linear relationship confirmed ✓

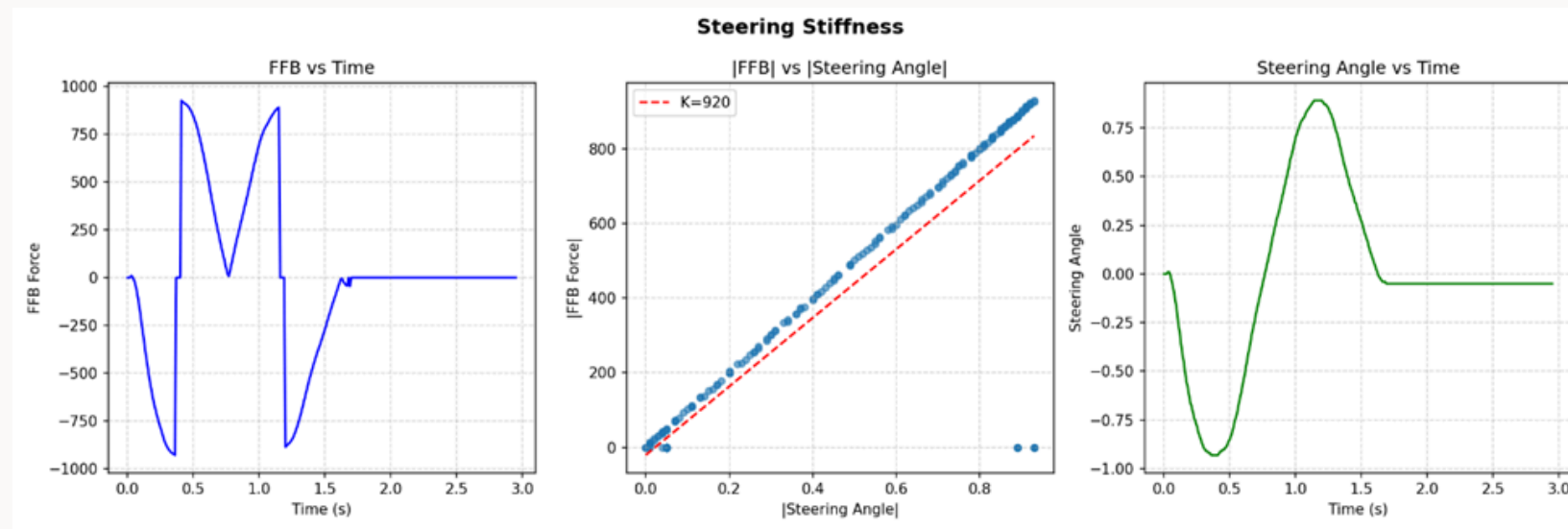
Friction Force



- Expected: High FFB during motion, near-zero at rest; decreases with speed
- Result: $T_{fric} \approx 1800$ at low speed, drops as speed_factor increases ✓

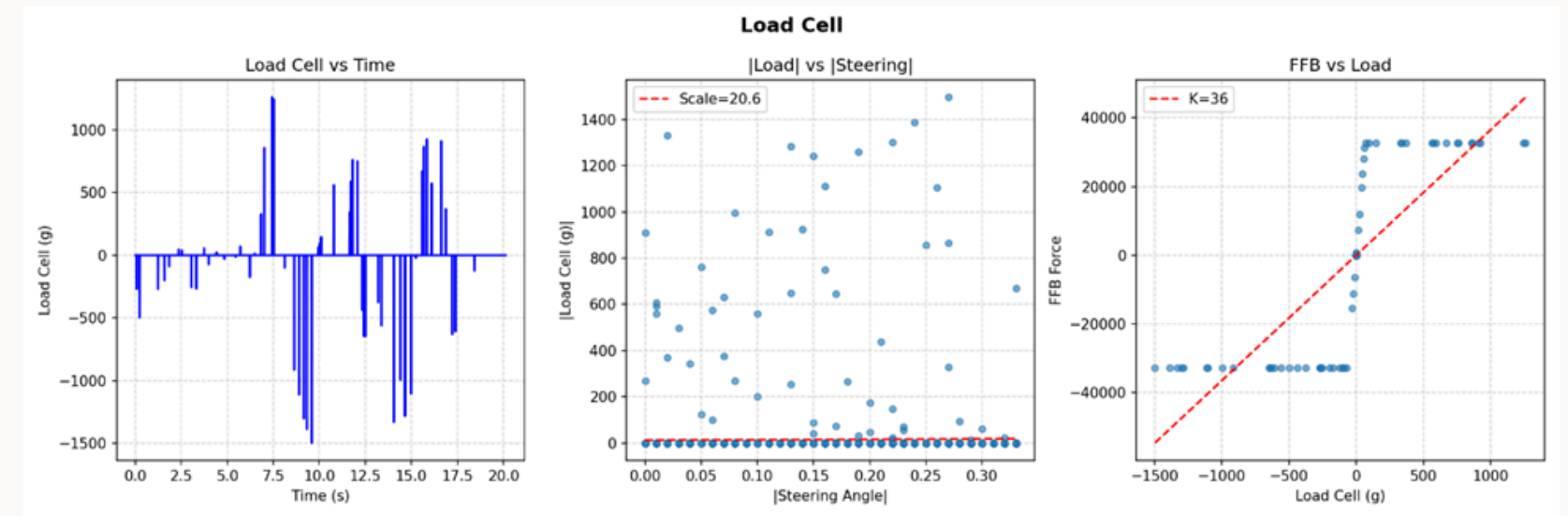
FFB Component Test Results

Steering Stiffness



- Expected: $|FFB| \propto |\theta|$, Higher FFB at Higher angle
- Result: FFB response as expected steering angle ✓

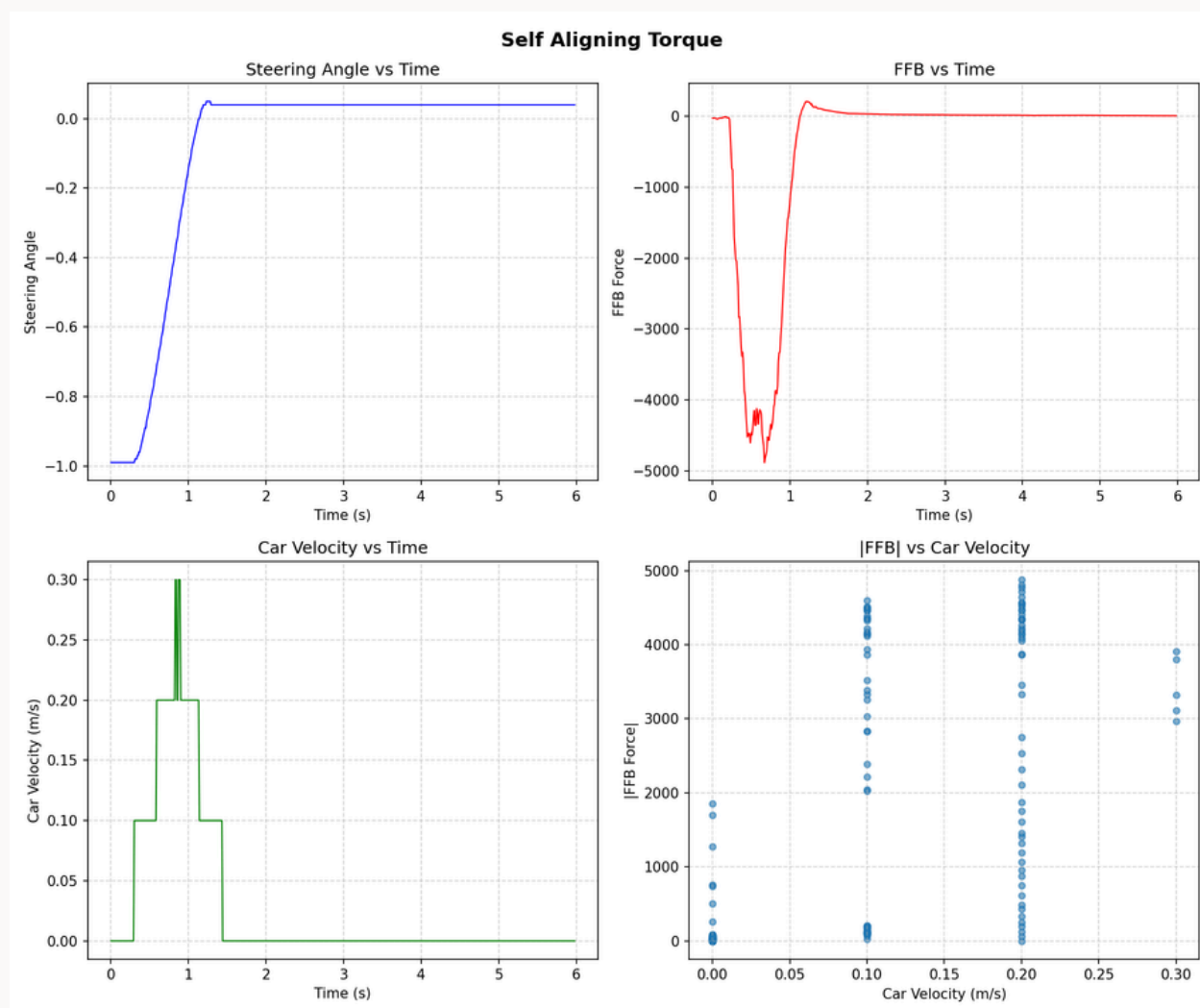
Loadcell



- Expected: FFB response to the load from load cell
- Result: FFB reaction as load cell get push ✓

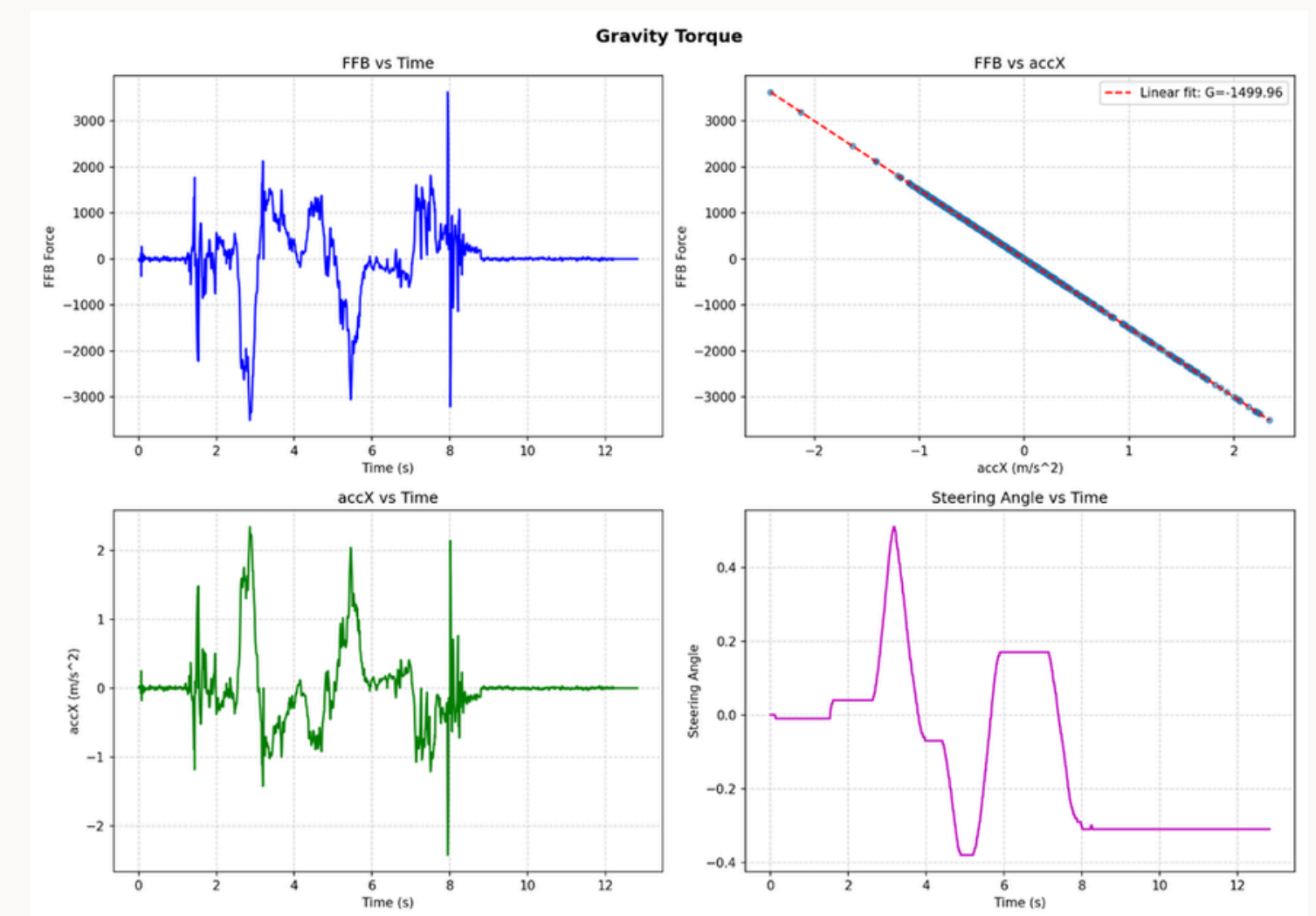
FFB Component Test Results

Self Aligning Torque



- Expected: $FFB \propto |\theta| \times \text{speed_factor}$, returns wheel to center
- Result: FFB increases with both angle and speed and wheel angle return to center ✓

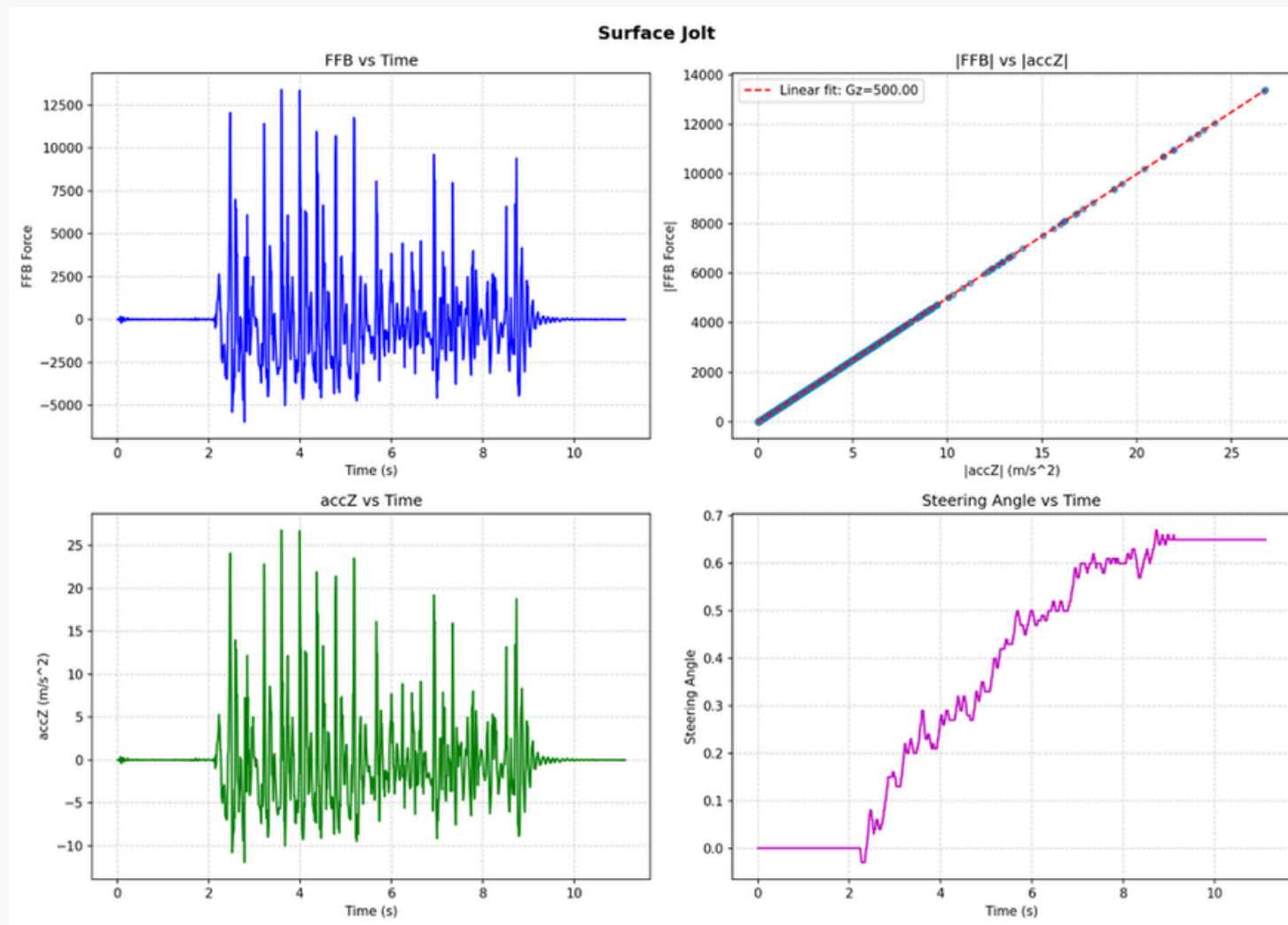
Gravity Torque



- Expected: $FFB \propto \text{accX}$, direction depends on tilt
- Result: FFB response to the correct tilt direction ✓

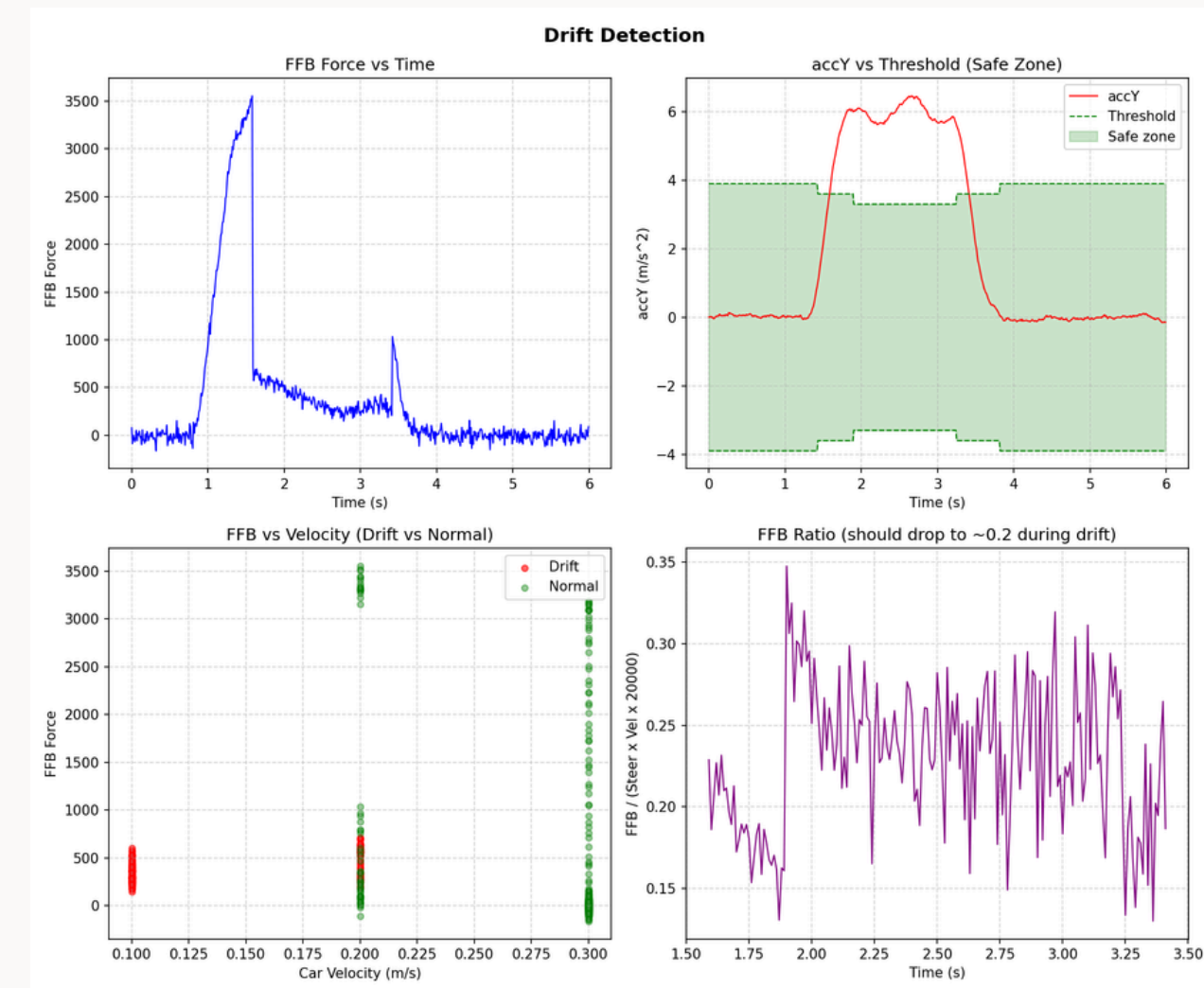
FFB Component Test Results

Surface Jolt



Expected: FFB with high-frequency response when car bump
Result: High frequency response and captures bump events ✓

Drift Detection

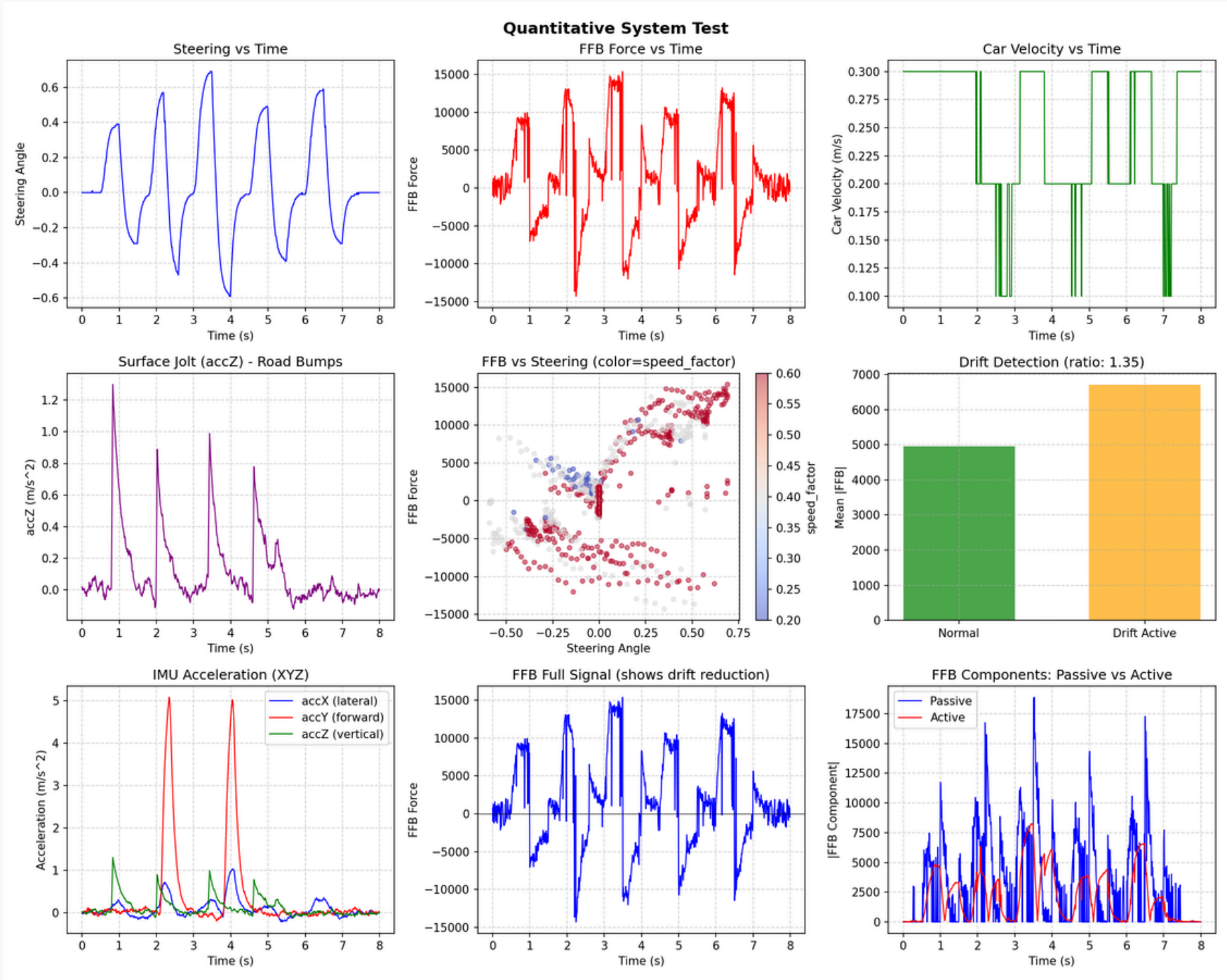


- Expected: FFB drops to ~20% when $|accY| > 3.0 \times (1 + |v|)$
- Result: FFB reduced by ~80% during drift, normal outside ✓



System Integration Test

FFB response to the right condition for each component (Damping, Friction, Stiffness, Load, SAT, Gravity, Jolt, Drift)



System Integration Test





Summary

- Latency avg. 5.20 ms — lower than 70 ms
- Range test 40 meters without packet loss
- Force Feedback system able to response based on vehicle behavior



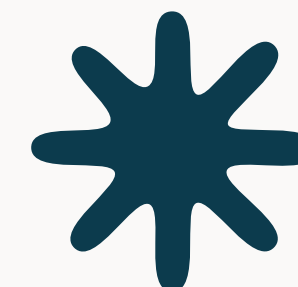
Recomendation & Future Work

Recomendation

- ACS712 not suitable (high noise)
- Improve Load Cell calibration (least-squares regression)
- Integration test is only in a limited space.

Future Work

- Integration test on more environment
- Add Adaptive Gain based on driver behaviour
- Add Heading Correction (yaw rate)
- Improve Drift Detection algorithm (adaptive)



THANK YOU

